

Ontology: Foundations

Abstract

The *Æther* / *Æther-flow* framework asks whether the geometry and temporal organization of physical experience can be understood in terms of a source structure that is not identified in advance with spacetime. Its present source assumptions are limited: a smooth four-dimensional manifold is selected, while the mathematical meaning of *Æther-flow* remains unspecified. General relativity is adopted separately as the effective gravitational benchmark. This paper distinguishes those assumptions from the proposed interpretation, explains what spatial experience and S-time would require a future reconstruction to establish, and states the conditions under which the effective model has the same predictions as general relativity. The framework is therefore an incomplete source proposal associated with a defined relativistic model, not a derivation of spacetime or evidence for a new material medium.

1 The question and the present answer

The *Æther* / *Æther-flow* framework is motivated by a question about physical explanation: could the spacetime relations described by relativity arise from a more basic organization, rather than being the starting point of the account? The words *Æther* and *Æther-flow* express that aim. They do not, by themselves, specify the objects or laws needed to answer it.

The present framework takes a conservative position on the measurable physics. It adopts general relativity (GR), with a single Lorentzian metric and specified ordinary matter, as the effective gravitational model. The source proposal is kept separate. This separation allows familiar relativistic calculations to be used without implying that the proposed source already explains their origin.

There are consequently two different questions. One is whether the adopted effective model reproduces the observations that the same GR model reproduces. Conditional on identical matter, parameters, solutions, and measurement rules, it does. The other is whether the source assumptions produce that effective model. No construction establishing the second statement is supplied here. The distinction is substantive: agreement obtained by adopting a theory is not evidence that its equations have been derived from another theory.

This paper establishes the vocabulary and the scope of that distinction. Dynamics states the effective action; Consistency examines its local gravitational degrees of freedom; Relativistic Recovery derives representative measurement relations; Geometry explains the observer-congruence language used to interpret them. Those companion analyses concern GR unless a source-to-target connection is explicitly supplied.

2 Motivation without a literal medium

The motivating picture treats change and spatial organization as related aspects of a possible underlying structure. In its historical form, the picture used phrases such as “intrinsic ordered motion,” “experienced space,” and “S-time.” These phrases identify questions worth making precise. They are not yet mathematical consequences of the selected source assumptions.

In particular, the proposal is not a return to a three-dimensional wind through which light propagates. Such a picture would need a physical velocity, a rest frame, interactions with measuring instruments, and quantitative consequences. None of those quantities is defined at source level in the present formulation.

Rejecting a naive wind picture does not establish an alternative microscopic theory; it only prevents an inappropriate interpretation of the current symbols.

It is also important not to confuse this use of Æther with Einstein-aether theory. The latter introduces a dynamical unit timelike vector with an action and additional coupled modes (Jacobson & Mattingly, 2004; Jacobson & Mattingly, 2001). The effective model here contains no corresponding independent gravitational vector. An observer's four-velocity will appear in the geometric description, but choosing observers is not the same operation as adding a field equation for a new physical substance.

3 The source assumptions

Definition 1 (Source arena). *The symbol $\mathcal{A}$ denotes a smooth four-dimensional manifold. Its manifold structure and dimension are assumed. No particular topology is selected by this statement, and no Lorentzian metric, causal order, physical clock, or matter content is assigned to it.*

The source text describes these assumptions as *primitive debt*: they are structure used at the beginning, not structure recovered by a derivation. Here that phrase means only “assumed rather than explained.” A later proposal may justify or replace these assumptions, but they cannot currently be advertised as consequences of Æther-flow.

Definition 2 (Unspecified source structure). *Φ_{src} is a placeholder for a possible source-order or evolution structure. Its mathematical type has not been fixed. It is not yet a map, vector field, differential operator, stochastic process, group action, or physical-time parameter.*

The distinction between a placeholder and a mathematical flow matters. A conventional flow might be specified by maps Φ_λ obeying a composition law and defined on a parameter domain. Writing such maps here would add assumptions about that domain, composition, and perhaps invertibility. The source definition explicitly leaves those choices unresolved. We therefore retain Φ_{src} without assigning it a convenient but unsupported type.

The source package can be summarized as

$$\mathcal{A} \quad \text{smooth}, \quad \dim \mathcal{A} = 4, \quad \Phi_{\text{src}} \quad \text{not yet typed.} \quad (1)$$

This is a statement of the current setup, not a closed source theory. It supplies no state space for evolving degrees of freedom, no admissible variations, no dynamics, no observables, and no map to clocks or detectors. These omissions constrain what can be inferred; they do not establish that every future specification will fail.

4 Source arena and physical spacetime

Let M denote the effective spacetime manifold and $g_{\mu\nu}$ its metric. Writing (M, g) separately from $\mathcal{A}$ prevents an ambiguity present in informal descriptions of the ontology. Both arenas may be four-dimensional, but equal dimension does not identify them. Nor does smoothness supply a metric or a causal cone.

A source-to-target reconstruction would need to explain how some independently defined source data determine, approximate, or otherwise represent (M, g) and its operational content. Whether this requires a map between manifolds, a map between state spaces, a quotient, or a limiting construction depends on the source model. No one of these possibilities is adopted here.

It follows that a familiar GR congruence u^μ cannot simply be declared equal to Φ_{src} . The former is a

well-defined vector field on an already Lorentzian spacetime; the latter has neither a type nor a target map. The geometric use of the word flow here is therefore an effective description of observer motion, not a construction of source motion.

A further distinction concerns reality claims. Selecting $\mathcal{A}$ as a mathematical arena does not establish that a physical substrate with that structure exists. Its proposed physical role is an ontological hypothesis. The effective metric, in contrast, already has an operational role within the adopted model: it enters the rules used to compare clocks, trajectories, and light signals. A future bridge must connect these roles rather than relying on the common use of geometric language.

5 What “experienced space” would need to mean

The phrase *experienced spatial slice* originally expressed the aim of explaining why observers describe local measurements using three spatial directions. GR already provides a precise local construction for that purpose. Given a future-directed four-velocity normalized by $u^\mu u_\mu = -c^2$, define

$$h_{\mu\nu} = g_{\mu\nu} + \frac{u_\mu u_\nu}{c^2}. \quad (2)$$

The projector $h^\mu{}_\nu$ selects vectors orthogonal to u^μ . Its image is a three-dimensional local rest space. This is an observer-dependent tangent-space statement, not a new source derivation (Ellis & van Elst, 1999).

Three different ideas must not be merged. A rest space specifies local spatial directions. A spacelike hypersurface organizes events into a spatial surface when the required integrability conditions hold. A visual scene is reconstructed from received signals and need not correspond to events simultaneous in that rest frame. The ontology has not shown that any of these is an embedded slice of $\mathcal{A}$.

For a smooth timelike congruence, vanishing vorticity gives local hypersurface orthogonality. A global slicing needs additional global conditions. The present framework does not assume that every observer family defines a global three-space, and it does not derive a source embedding $\Sigma \hookrightarrow \mathcal{A}$. A future account of experienced space must state which of these operational or geometric targets it aims to reconstruct.

6 S-time: order, duration, and experience

S-time is retained as the name of an interpretive aim: relating temporal experience to the organization of change. No S-time functional is defined by the present source package. In particular, no monotonicity law or calibration identifies source order with measured duration.

Order and duration are different. An order may distinguish which events can precede others without assigning a number of seconds to an interval. In relativity, proper time assigns duration along a timelike trajectory using the metric. A thermodynamic arrow introduces further questions about irreversible behavior and boundary conditions. Subjective temporal experience raises questions not settled by the mere existence of a clock variable. The source manuscripts contain no derivation connecting all these notions.

The effective model therefore uses ordinary proper time. For a timelike worldline Γ ,

$$\tau[\Gamma] = \frac{1}{c} \int_\Gamma \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}. \quad (3)$$

This operational prescription is adopted from GR (Carroll, 1997). Calling it an expression of S-time is an interpretation only until a source construction reproduces the same clock readings and their dependence on

trajectory. No universal preferred clock or source time is inferred from the notation.

7 The adopted relativistic benchmark

7.1 Fields, action, and conventions

The effective arena is a time-oriented four-dimensional Lorentzian spacetime with signature $(-+++)$ and Levi-Civita connection. Greek indices run from 0 to 3. In the covariant action, coordinates are length-valued, with $x^0 = ct$; S_m has the units of physical action. The benchmark is

$$S_{\text{eff}}[g, \psi] = \frac{c^3}{16\pi G} \int_M d^4x \sqrt{-g} (R - 2\Lambda) + S_m[g, \psi], \quad (4)$$

where $G > 0$, Λ is constant, and ψ denotes specified matter fields. Taking $\Lambda = 0$ gives the original noncosmological specialization. Matter is coupled to the same metric, with ordinary locally Lorentz-invariant matter laws when the standard light and clock interpretations are used.

With the physical-action stress tensor

$$T_{\mu\nu} = -\frac{2c}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}, \quad (5)$$

the bulk field equation is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (6)$$

Dynamics gives the variation and its boundary assumptions. No source variable occurs in this action as a new propagating gravitational field.

7.2 Operational content and equivalence

The effective metric supplies the causal classification of tangent directions. Standard Maxwell light in vacuum geometric optics follows null directions,

$$g_{\mu\nu} k^\mu k^\nu = 0, \quad (7)$$

and ideal clocks measure (3). Neutral structureless test bodies follow timelike geodesics when nongravitational forces and finite-size effects can be neglected. These are model assumptions and standard consequences, not new laws induced by an already defined source flow.

The term *exact closure* names the decision to fix this effective gravitational model to GR. For a specified matter model, the two descriptions have identical observables only when they also use the same constants, admissible solutions, initial and boundary data, and measurement rules. If a source interpretation imposed an additional restriction on possible solutions, that restriction would have to enter the comparison. Equality of the printed metric equation alone would no longer suffice.

The action with unspecified S_m thus defines a family of possible GR models. It does not select one cosmological history or one laboratory outcome. Its purpose is to provide a reference against which a completed source model could be compared.

8 Expansion, light, and the limits of interpretation

The motivating ontology includes expansion, light propagation, and temporal change. In the adopted benchmark, cosmological expansion is described by a changing FLRW scale factor and the expansion of comoving worldlines. No external container is needed for that description. Light propagation and gravitational redshift follow the same metric and observer measurements developed in Relativistic Recovery.

These facts provide a vocabulary for the proposed interpretation, not evidence that Φ_{src} produces them. A comoving expansion scalar $\theta = 3H$, for example, is calculated after a spacetime metric and congruence have been supplied. It cannot serve as the missing source dynamics without reversing the intended explanatory direction.

The same caution applies to metaphor. Static support, rotational observer motion, and wave-induced deformation can all be described using congruences, but no single image of a vortex exhausts gravity. Geometry makes these distinctions quantitative. Its quantities depend on the chosen observers as well as the metric and are not separate manifestations of an independently established source field.

9 How a later source proposal should be assessed

A useful source development would first define the objects left open here. It would then state a particular reconstruction claim and show which source assumptions imply it. The claim might concern a causal structure, an operational clock, a metric, or a field equation; these are different levels of achievement. Recovering one does not automatically recover the others.

A successful bridge must avoid importing its intended conclusion as a disguised premise. Starting with a Lorentzian source metric and identifying it with g may define a model, but it would not derive Lorentzian geometry from the present metric-free source assumptions. Similarly, postulating the Einstein–Hilbert action under a different name is not a derivation of that action. Such choices can be legitimate assumptions if they are acknowledged as assumptions.

If an optional extension fails a consistency or observational test, the independent GR benchmark remains a reference. This is a research policy, not a theorem protecting all source hypotheses. A failure may provide evidence against the source assumptions responsible for it. Conversely, absence of a bridge in these manuscripts is not a proof that no bridge can be constructed.

10 Conclusion

The present ontology consists of a selected smooth four-dimensional source arena, an untyped Æther-flow placeholder, and an interpretive aim concerning spatial and temporal organization. Its quantitative effective model is GR with specified ordinary matter. The geometric and operational results developed in the companion papers belong to that adopted model.

This division preserves the source question without overstating its answer. The existing assumptions do not establish a physical substrate, a source clock, or a derivation of Einsteinian dynamics. They define where further explanation is sought. A future result must supply the missing source structure and demonstrate a specific bridge; until then, exact agreement with GR is a property of the adopted benchmark rather than independent confirmation of the ontology.

References

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